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① A Carmichael number is a composite number n such that for every integer a relatively prime to n , it satisfies:

$$a^{n-1} \equiv 1 \pmod{n}$$

$$1729 = 7 \times 13 \times 19$$

all three are primes and distinct and:

$$7-1=6$$

$$13-1=12$$

$$19-1=18$$

check if 1729 satisfies Korselt's Criterion

1. n is square-free

2. $p-1 \mid n-1$ for every prime $p \mid n$

$$6 \mid 1728$$

$$12 \mid 1728$$

$$18 \mid 1728$$

So, 1729 is a Carmichael number.

② primitive Root of $\mathbb{Z}_{23}$

Ans: an Integer g such that powers $g^1, g^2, \dots, g^{22}$ mod 23 give all non-zero residues mod 23

Euler's totient $\phi(23) = 22$, so we want g such

$$\text{Ord}_{23}(g) = 22$$

They $g=5$:

Compute powers of 5 mod 23

$$5^1 \text{ mod } 23 = 5 \Rightarrow \text{Not } 1$$

$$5^2 = 25 \text{ mod } 23 = 2 \neq 1$$

All lower powers not giving 1 $\Rightarrow 5$ is a primitive root modulo 23.

③ Is $\langle \mathbb{Z}_n, +, \times \rangle$ a ring?

$\rightarrow \mathbb{Z}_n$ is the set of integers modulo n

- Under addition and multiplication mod n

To be a ring, the set must:

1. be an abelian group under addition and multiplication

2. Have multiplication associative

3. Multiplication distributes over addition

4. closure under multiplication

$\mathbb{Z}_n$ have all these characteristics so, $\mathbb{Z}_n$ is a ring.

④ Are $\langle \mathbb{Z}_{37}, + \rangle$, $\langle \mathbb{Z}_{35}^*, \times \rangle$ abelian groups?

Ans:

Addition mod 37

→ $\mathbb{Z}_{37}$ is a finite field (since 37 is prime)

- Additive group is always abelian,

So, $\langle \mathbb{Z}_{37}, + \rangle$ is abelian group.

$\langle \mathbb{Z}_{35}^*, \times \rangle$:-

→ set of unit modulo 35

→ $35 = 5 \times 7$ so not prime $\Rightarrow$ not a field

→ units mod 35 are integers $\langle 35 \rangle$ that are coprime to 35.

→ $\phi(35) = 24 \Rightarrow$ order 24

So, $\mathbb{Z}_{35}^*$ under multiplication is an abelian group.

⑤ We want to construct that finite field

$\text{GF}(2^3)$

• Base field : $\text{GF}(2) = \{0, 1\}$

• Degree = 3

• pick an irreducible polynomial of degree 3 over $\text{GF}(2)$

Example: $f(x) = x^3 + x + 1$

Now Define, $\text{GF}(2^3) = \text{GF}(2)[x] / (x^3 + x + 1)$

Element of $\mathbb{GF}(2^3)$ are all polynomials of degree < 3 with coefficients in $\mathbb{GF}(2)$.

So, the elements are?

$$\{0, 1, x, x+1, x^2, x^2+1, x^2+x, x^2+x+1\}$$

There are $2^3 = 8$ elements.

Arithmetic: Addition - is Done modulo 2
(bitwise XOR)
Multiplication is Done modulo x^3+x+1

Example:

$$\text{Let, } a(x) = x^2+1, b(x) = x+1$$

Then,

$$a(x) \cdot b(x) = (x^2+1)(x+1) = x^3 + x^2 + x + 1$$

Now, reduce mod x^3+x+1

$$x^3 + x^2 + x + 1 \equiv (x^3 + x + 1) + x^2 + x + 1 = x^2$$

$$\text{So, } (x^2+1)(x+1) = x^2 \text{ mod } (x^3+x+1)$$